Kabir Rohit Khanna

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Education

Oct 2022 – Present (Expected Jun 2023)

University of Oxford *Oxford, UK*
MSc. Mathematical and Theoretical Physics
Coursework:
- Quantum Field Theory
- Advanced Quantum Theory
- Groups and Representations
- Topological Quantum Theory
- Introduction to Quantum Information†
- Quantum Matter 1†
- Quantum Matter 2†

Aug 2017 – May 2022

Indian Institute of Technology Madras *Chennai, Tamil Nadu*
B.Tech in Engg. Design + M.Tech in Quantum Science & Technology (offered by EE and PH)
CGPA: 9.41/10 - Rank: 3/57
Relevant coursework: Grade Scale: 10
- General Relativity and Cosmology (10)
- Q. Computation and Information (10)
- Mathematical Physics 2 (9)
- Experimental Techniques for Quantum Computation and Metrology (9)
- Quantum Electronics and Lasers (9)
- Stochastic Processes (10)
- Optical Signal Processing and Quantum Communication (10)

Aug 2019 - Dec 2019

Technical University of Denmark (DTU) *Lyngby, Denmark*
Exchange Student - Semester 5
Relevant coursework:
- Statistical Physics
- Advanced Fluid Mechanics
- Discrete Mathematics 2: Algebra
- Signals and Systems in Discrete Time

Publications

[1] **Kabir Khanna** and Saurya Das, *GUP Corrections to the Jaynes-Cummings Model*, arXiv (2022). https://doi.org/10.48550/arXiv.2209.10152

[2] [*In preparation*] Raghvendra, Singh, **Kabir Khanna**, and Dawood Kothawala, *Decoherence by Time Dilation in Curved Spacetimes.*

Research Experience

Aug 2021 - May 2022

IIT Madras *Chennai, India* – **Master's Thesis** — *Grade: 10/10*
Supervisor - Prof. Dawood Kothawala
Title: *Aspects of Quantum Information in Curved Spacetimes*
- *Part 1* - Decoherence due to Time Dilation in Curved Spacetimes.
- *Part 2* - A pedagogical introduction to the BH information paradox.

Jun 2021 - Sep 2022

University of Lethbridge *Remote* – **Research Internship**
Supervisor - Prof. Saurya Das
Topic: Intersection of quantum gravity phenomenology and quantum optics.

†To be tentatively taken later in the course

Jan 2021 - Jun 2021 **Robert Bosch GmbH** *Bangalore, India* – **R&D intern in Quantum Machine Learning**

Topic: Applying NISQ algorithms like QAOA to real use case problems to look for potential speedups and gauge the efficacy of quantum algorithms in the industry.

Jun 2019 - Jul 2019 **Wolfram Summer School** *Waltham, Massachusetts* – **Summer Research Student**

Supervisors - Christopher Wolfram and Jonathan Gorard

Topic: Aggregation Systems

- Contribution to the Wolfram Function Repository can be found here.
- A short write-up can be found here.

Awards and Recognitions

2022 OCSI Scholarship awarded by the Oxford Cambridge Society of India.

2021 Selected[§] for the DAAD WISE Research Scholarship.

2019 Selected to participate in a tuition-funded semester exchange at DTU, Denmark.

2019 Awarded partial funding to attend the Wolfram Summer School at Waltham, Massachusetts.

2018 Awarded the best volunteer at NSS IITM.

2017 Silver medal for obtaining a top 4000 rank in JEE Advanced 2017 out of 1.2 million students.

2015 Awarded the best outgoing student at my school, DPS Secunderabad.

Tutoring and Volunteering

Jun 2022 - Present **Freelance Math Tutor at Gauthmath**

- Mathematics expert at Gauthmath, an app that helps students with homework problems.

Jan 2022 - May 2022 **Teaching Assistant for PH3520 Quantum Physics** *IIT Madras, Chennai*

Feb 2021 - May 2021 **Academic Mentor at Saathi** *IIT Madras, Chennai*

- Chosen as a part of a Saathi initiative to academically guide three first year students through their initial virtual semesters.

May 2018 - Jul 2018 **Mathematics and Physics Tutor** *Lamdon School, Leh*

- Tutored over 200 high school and middle school students of a remote town near the Himalayas in mathematics and physics.

July 2017 - May 2018 **Mathematics Educator at NSS** *IIT Madras, Chennai*

- Volunteered to present mathematical concepts through props and presentations to marginalized middle school students in Chennai.

Skills

Languages/Tools: Mathematica, LaTeX, Python, C, MATLAB

Quantum Computing Libraries: Pennylane, Qiskit, Braket, Cirq

[§] I Did not accept it since I had already begun working with Prof. Saurya Das when the DAAD WISE results were announced.